

Forecasting 21-Day Realized Volatility of the S&P 500: A Comparison of Econometric and ML Models

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Abstract

I compare out-of-sample forecasts of 21-day realized volatility of the S&P 500 index from **five** models: naive persistence, a HAR-style ¹ regression, GARCH ², ordinary least squares and support vector regression. The ranking depends on the loss function: GARCH has the lowest QLIKE loss but barely improves on persistence in mean squared error, while the regression models achieve the highest R^2 but QLIKE losses close to persistence.

1 Question and target

The question is whether machine-learning regressions improve out-of-sample volatility forecasts relatively to persistence and classical conditional-volatility models.

Let $r_t = \log(P_t/P_{t-1})$ be the daily log return of the S&P 500 index ($\sim$ GSPC, daily closes from Yahoo Finance). **The forecast target at date t is the annualized volatility of the next 21 daily returns**

$$RV_{t+1:t+21}^{(21)} = \sqrt{252} \text{sd}(r_{t+1}, \dots, r_{t+21}), \quad (1)$$

where sd is the sample standard deviation. The predictor $RV21_t$ is the same quantity computed on $r_{t-20}, \dots, r_t$, so predictor and target windows never overlap.

One property of this design to keep in mind: consecutive targets share 20 of their 21 returns, so daily observations are strongly dependent; a naive evaluation on every day leak future information into training.

2 Evaluation design

Walk-forward with embargo. Starting from an initial sample of 1,000 trading days, the model is fitted and used to predict the next block of 21 trading days; the origin then advances by 21 days and the model is refitted on the expanded sample. Because the training target at date s uses returns up to $s + 21$, the training set is truncated 21 days before the first test date (the *embargo*), so that no training target overlaps the test block (Figure 1). All preprocessing (feature standardization) is fitted on the training sample only.

¹Heterogeneous Autoregressive model of Realized Volatility. Canonical **HAR-RV**, as Corsi[3] described it, assumes that tomorrow's volatility is a linear function of past volatilities across different horizons. In our project, however, rv_{21} already represents a 21-day realized volatility. We therefore adapted a "HAR-RV-style" to forecast the next 21-day realized volatility using volatility information using volatility on different horizons. More information in the notebook.

²Generalized Autoregressive Conditional Heteroskedasticity. Model is briefly described in the notebook

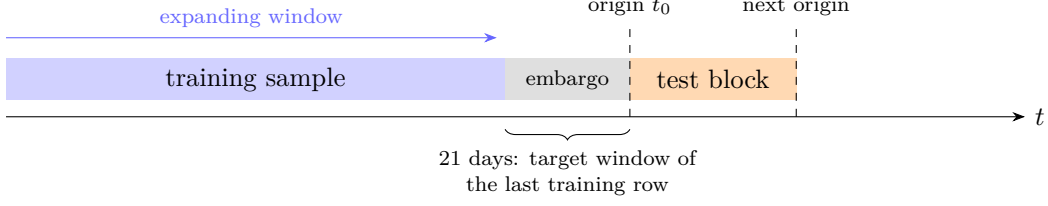


Figure 1: Walk-forward scheme. The model is refitted every 21 trading days on all observations whose target window ends before the forecast origin t_0 .

Selection and test periods. The walk-forward forecasts are split at SPLIT date = 01/01/2017. Forecasts before this date form the *selection period*, used for diagnostics and for choosing the SVR hyperparameters. Forecasts after it form the *final test period*, which is used only once, for the results reported in Section 4.

Loss functions. The primary metric is the QLIKE loss computed on variances,

$$\text{QLIKE} = \frac{1}{N} \sum_t \left[\frac{\text{RV}_t^2}{\hat{\sigma}_t^2} - \log \frac{\text{RV}_t^2}{\hat{\sigma}_t^2} - 1 \right], \quad (2)$$

QLIKE is asymmetric: it penalizes under-prediction of volatility much more heavily than over-prediction, and it is invariant to the overall scale of volatility. I also report RMSE, R^2 , the correlation between forecasts and realizations, and the out-of-sample R^2 relative to a benchmark b ,

$$R_{\text{vs } b}^2 = 1 - \frac{\sum_t (\text{RV}_t - \hat{\sigma}_t)^2}{\sum_t (\text{RV}_t - \hat{\sigma}_t^b)^2}, \quad (3)$$

with b equal to the naive or the HAR-style forecasts.

3 Models

The models form a ladder of increasing flexibility; each must beat the ones below it to justify its complexity.

1. **Naive persistence.** $\hat{\sigma}_t = \text{RV}_{21t}$: next month's volatility equals last month's.
2. **HAR-style regression** [3]. OLS on volatility components at three horizons, all known at t :

$$\text{RV}_{t+1:t+21}^{(21)} = \alpha_0 + \alpha_1 \sigma_t^{(1)} + \alpha_2 \sigma_t^{(5)} + \alpha_3 \text{RV}_{21t} + \varepsilon_t, \quad (4)$$

with $\sigma_t = \sqrt{252} |r_t|$ and $\sigma_t^{(5)}$ its 5-day average. Since only daily closes are used, $|r_t|$ stands in for the intraday realized volatility of the original HAR-RV model.

3. **manually implemented GARCH(1,1)** [2] I tried to manually implement a GARCH model for educational purposes, which was then compared with an existing function in arch library.
4. **Linear regression (OLS)** on seven standardized features: RV_{21t} , its lags at 1, 5 and 21 days, price relative to its 20- and 50-day moving averages, and a 14-day RSI.
5. **Support vector regression** on the same features. The kernel, C and γ are selected by QLIKE on the selection period among 3 configurations; the selected one is **linear kernel**, with $C = 10$. The ε -insensitive tube is set to 0.01, i.e. small relative to the level of annualized volatility (around 0.15).

4 Results

Table 1 reports the final-test-period results.

Model	QLIKE	RMSE	R^2	$R^2_{\text{vs naive}}$	$R^2_{\text{vs HAR}}$	Corr
GARCH(1,1)	0.661	0.1044	0.141	-0.015	-0.097	0.586
HAR-style	0.754	0.0997	0.217	0.075	0.000	0.592
OLS	0.822	0.0962	0.271	0.139	0.069	0.598
Naive	0.851	0.1036	0.153	0.000	-0.081	0.578
SVR (selected)	0.861	0.0956	0.279	0.149	0.080	0.596

Table 1: Out-of-sample results on the final test period (02/01/2017 up to 01/01/2024), . Models sorted by QLIKE ascending order.

The ranking depends on the loss. GARCH(1,1) has the lowest QLIKE, yet its squared error is essentially that of naive persistence ($R^2_{\text{vs naive}} = -0.015$). Conversely, SVR and OLS have the lowest RMSE and the highest R^2 , but QLIKE values close to persistence (SVR is in fact the worst of the five on this metric). All five models have almost the same correlation with realized volatility (0.58–0.60): they contain similar information and differ mainly in the *level and asymmetry* of their errors.

This pattern is coherent with the asymmetry of QLIKE. Regression models are fitted by least squares, which is optimal for MSE by definition but tolerates occasional large under-predictions. QLIKE, on the other side, punishes these underpredictions.

Loss comes mainly from COVID underpredictions: A small number of dates carries most of the QLIKE loss. The largest single error of the test period is the origin immediately preceding the Covid crash, where realized volatility of 0.79 was forecast at roughly 0.12 by every model.

Calibration. Figure 3 plots realized against forecast volatility with the regression $RV_t = a + b \hat{\sigma}_t$ [5]; a perfectly-calibrated forecast has $a = 0$ and $b = 1$. All models have $b < 1$, which can be interpreted as over-reaction

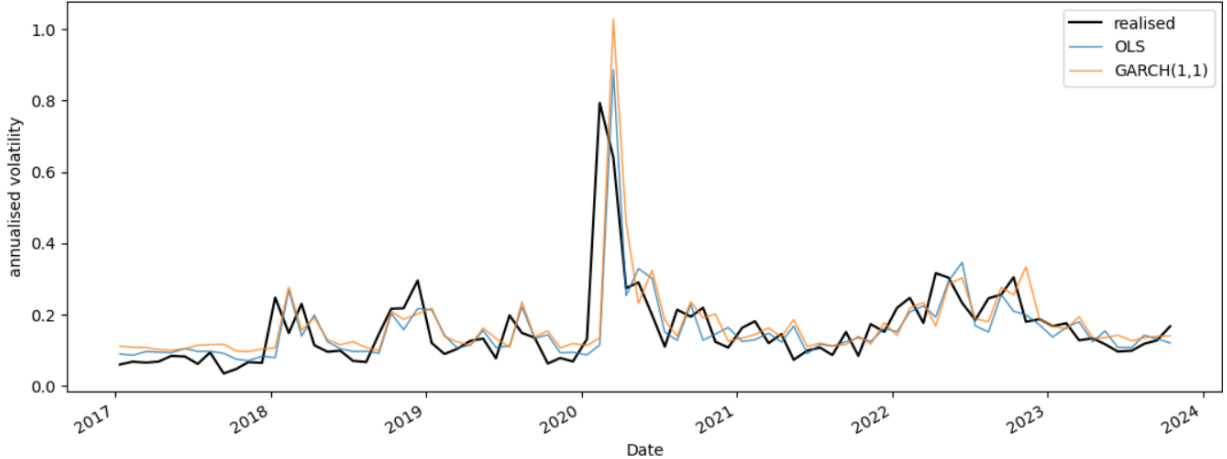


Figure 2: Forecast and realized 21-day volatility over the final test period.

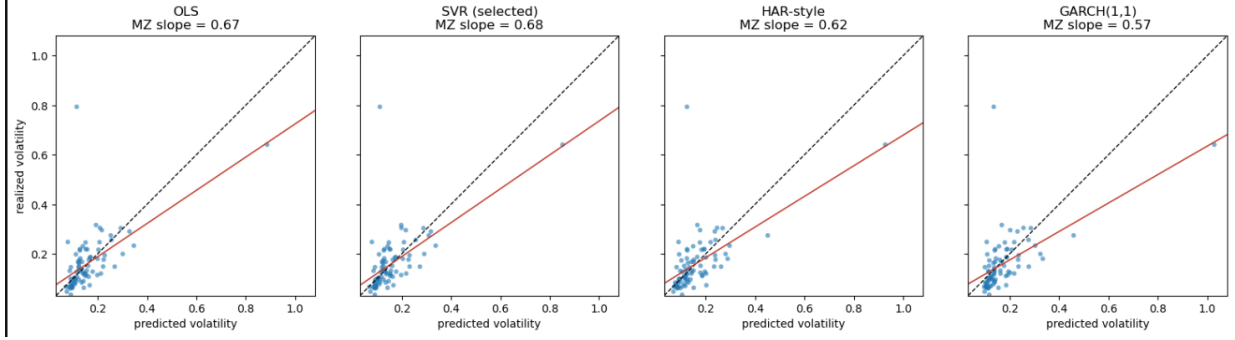


Figure 3: Realized versus forecast volatility, one panel per model. Dashed: perfect forecast. Red: Mincer–Zarnowitz regression line.

5 Implementation notes

GARCH estimation. GARCH(1,1) was first implemented by hand and checked against the `arch` package.

Horizon (days)	Manual	<code>arch</code> lib	Relative diff. (%)
1	0.4360	0.4424	1.44
2	0.4532	0.4588	1.23
3	0.4700	0.4749	1.04
4	0.4865	0.4907	0.86
5	0.5027	0.5062	0.70

Table 2: Conditional variance forecasts from the hand-written GARCH(1,1) estimator and from the `arch` package.

Leakage checks. Every predictor at date t uses returns up to t only; every target uses returns from $t + 1$ on. The embargo guarantees that the last training target ends before the forecast origin.

6 Limitations and extensions

- **Realized volatility.** Daily close-to-close returns give a noisy 21-day volatility estimate. Intraday realized volatility [1] would reduce the noise in both the target and the HAR components, and is the setting the HAR model was designed for.
- **Sample size.** The test period yields fewer than 90 independent forecasts; conclusions about the ranking should be read in that light.

7 Conclusion

The five models cannot be separated on this sample. GARCH(1,1) has the lowest QLIKE, but no pairwise Diebold–Mariano test comes close to significance: $p = 0.26$ against HAR, 0.33 against OLS, 0.32 against SVR, and 0.10 against naive persistence. Eighty-two forecasts are not enough to rank these models, and the squared-error ordering is no firmer.

The ranking also depends on which loss is used. GARCH wins on QLIKE but does not beat persistence on squared error ($R_{vs\ naive}^2 = -0.015$); OLS and SVR reverse both. All five correlate with realized volatility at 0.58–0.60, so they carry much the same information and differ in how their errors are distributed. Least squares tolerates occasional large under-predictions, which is what QLIKE punishes hardest. GARCH reacts to shocks within a day and so avoids those, but over-predicts once volatility falls back, which squared error punishes instead.

The 2020 Covid episode shows how little it takes to move these numbers. The worst error in the test period is the origin just before the crash: realized volatility near 0.79 against a forecast of about 0.12. QLIKE grows with the ratio of realized to predicted variance, so that one date carries a large part of every model’s total loss.

So the result is negative. On daily closing prices at a 21-day horizon, none of these machine-learning specifications beats HAR or GARCH by a margin that survives a test, and the apparent ordering turns on the choice of loss and on a few dates. Settling the question would need a cleaner volatility proxy from intraday data, implied volatility in the feature set, and more than seven years of monthly forecasts.

References

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